

2. Full Trash Capture Devices

2.1 Capital Improvement Project Alternatives

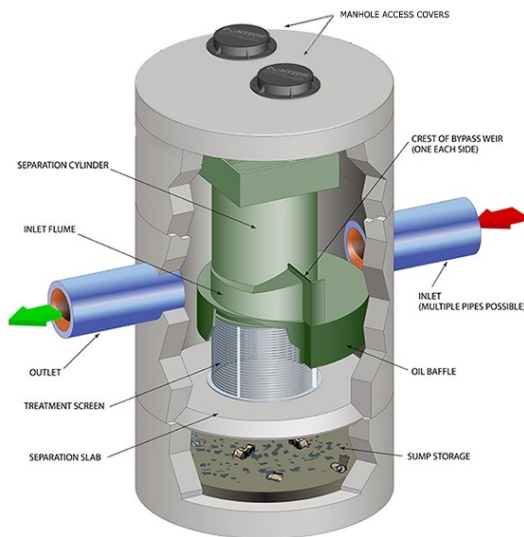
There are a variety of devices available to meet the full capture trash reduction goal, including:

- Catch basin inlet filters/screens;
- Netting, screens, or hydrodynamic structures on a storm drain trunk line; and
- End-of-pipe netting, screens, or hydrodynamic structures.

When selecting a device, it is important to consider the drainage area, hydraulic losses across the system when full, permitting requirements in open channels, device maintenance and access, property rights, and utility clearance. All of these factors should be weighed against the device, installation, and maintenance costs to ensure maximum capture volume per dollar spent. The current list of approved full trash capture devices is contained in Attachment 2.

2.1.1 Large Scale Devices

This study focuses on large scale devices as they have the capability of capturing runoff from Caltrans right-of-way and thus enabling possible funding sources. Large-scale devices provide a centralized location for construction and maintenance. The downside of large devices is that they capture flow from low trash generating areas resulting in the need to have a larger device to treat the additional runoff from areas with little trash reduction benefit.



Hydrodynamic separators (HDS) are widely used in stormwater treatment. They are flow-through structures with a settling or separation unit to remove sediments, floatables, and other pollutants. They come in a wide range and some are as small as a standard manhole structure. These inline systems can be cleaned using a vacuum truck to pump out the trash, sediment, and water that has collected in the bottom. Engineering and installation costs can be quite high, but the devices are long-lasting and can capture trash from a considerable area.

Netting devices can be placed on discharge pipes that will collect trash. Netting systems have one or more mesh bags and a metal frame guide system to support the nets. When full, the nets can be removed and replaced with new nets. End-of-pipe systems can be installed at the actual discharge point to receiving water, or just a collection point. Because these nets are installed above ground, they must be inspected at regular intervals to find and repair any damage. The netting devices may also require permitting for installation and maintenance if they are located in the channel.



3. Cost Estimates and Maintenance Considerations

3.1 Unit Costs

The following assumptions were made in determining construction costs for each proposed device:

- Installation costs for large-scale HDS and baffle devices are assumed to be 4 times the cost of the device;
- Installation of netting large-scale devices are assumed to be 3 times the cost of the device;
- Installation of louvered panels was calculated on a case-by-case basis as they are custom installations;
- Hydrodynamic separators and baffle boxes capital cost is estimated at \$3000 per treatment flow rate (cfs) based on recent bids;
- Netting and large screen device capital costs are estimated at \$2500 per treatment flow rate (cfs) based on recent bids;
- Small scale devices (i.e. inlet screens and hanging baskets) were estimated at \$2,500 per device, based on recent quotes;
- The devices are located in areas which do not require major utility relocation, property or easement acquisition or environmental permitting;
- Large scale devices were assumed to have a lifetime of 50-years;
- Small scale devices were assumed to have a lifetime of 25-years;
- Total Base Cost is presented in the 2022 construction year dollars and includes installation, construction needed to update drainage infrastructure to support trash capture device, and the device. It does not include maintenance or device replacement;
- 20% construction contingency added to total costs
- 15% design added
- Permitting costs were estimated for each specific site location.

Table 4-1 lists the unit costs used to develop each of the site's installation cost. Attachment 3 contains details of each site quantities and total cost.



Table 3-1: Unit Cost Assumptions

Item	Cost per Unit	Unit
HDS or Baffle	\$3,000	per cfs
Netting Device	\$2,500	per cfs
Louvered Panel	\$500	SF
Connector Pipe Screen	\$2,500	each
Permitting for Underground	\$27,000	lump sum
Permitting for in channel	\$105,000 - \$120,000	lump sum

3.2 Maintenance Considerations

Cleaning and maintenance costs are not included in the installation costs. They are provided in this section for reference only for the MCSTOPPP consideration. Maintenance of both small-and large-scale devices can be done by municipal staff or hired out to a cleaning contractor. In the Bay Area, there is a mix of jurisdictions that perform maintenance in-house and those that contract out the maintenance.

For cleaning, replacement, and maintenance, the following considerations should be taken:

- Large scale HDS or baffle box devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$23,400/device annually;
- Netting devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a crane truck at \$5,000/half day, replacement net at \$1,500, or \$27,300/device annually;
- Louvered panels at the Rossi pump station are cleaned approximately three (3) times a year, requiring 4 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$7,200/device annually;
- Inlet screens are cleaned approximately three (3) times a year, requiring 4 hours of a team of two maintenance workers at \$100/hour- or \$7,200/device annually;

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



Project ID: LS-028-Alt1						
Project Name: Ferry Terminal						
Project Location: City of Larkspur (37.9452, -122.5122) (Project drains Caltrans right-of-way area as well)						
Site/Device Information						
Proposed Device Type: Netting Devices (See page 2 for Alternatives)						
Project Description: Construction of an end-of-pipe netting system and concrete pad. The netting devices would be attached to the two 42-inch pipes discharging into the creek parallel to Larkspur Ferry Terminal Pathway.						
Site Specific Considerations: - Pipe outfalls have existing flap gates - Location is tidal - Channel is privately owned						
O&M Summary: May require creek drawdown for maintenance.						
Comments/Notes: GG transit owns the outfall and creek, so device installation would need to be coordinated. Permitting may present a challenge, however, upstream locations are problematic due to the high traffic along Sir Francis Drake. Upstream locations also miss Caltrans.						
Drainage Area and Trash Reduction Information						
Total Drainage Area: 59 acres - Larkspur – 54 acres - Caltrans Right-of-Way – 5 acres						
Priority Land Use (PLU) Areas (acres): 41 acres						
Public Agency	Trash Generation Category				Non-PLU Area	Total Area
	Low	Mod	High	Very High		
Larkspur	12	29	0	0	13	54
Caltrans	--	--	--	--	--	5
1-yr/1-hr Peak Flow (modeled): 24 cfs						



Preliminary Cost Estimate	
Design/Permitting	\$173,000
Construction	\$241,000
Construction Contingency	\$48,300
Total Capital	\$523,000
Annual Operation & Maintenance	\$27,300/yr
Construction Cost per Acre of PLU	\$8,500/ac

